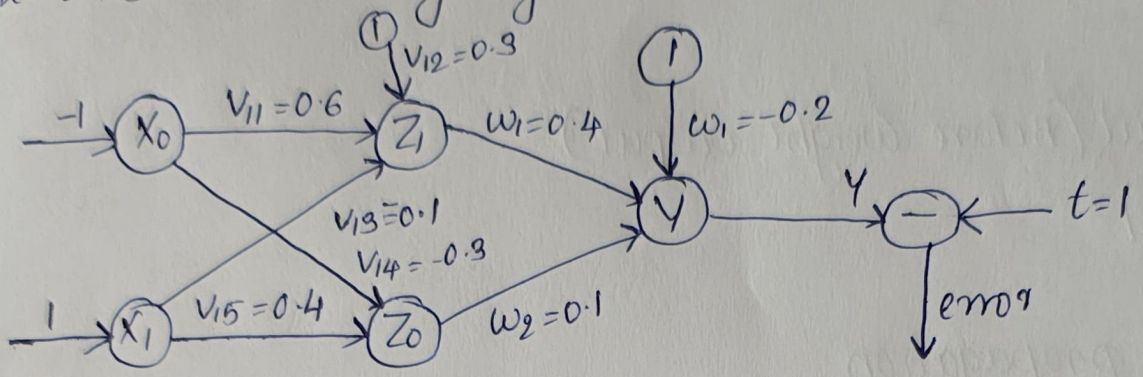
Using the back-propagation network, find the new weights for the network shown in the figure. It is presented with the input pattern [-1, 1] and the target output is 1. Using a learning rate $\alpha = 0.25$ and binary sigmoidal activation function.



1. Initial weights

weight : bias → $Z1 = 0.3$

bias → $Z_2 = 0.5$

2 Forward propagation

Hidden layer

For Z_1

$$net\ Z_1 = (-1)(0.6) + (1)(-0.1) + (1)(0.3)$$

$$net\ Z_1 = -0.6 - 0.1 + 0.3 = \underline{\underline{-0.4}}$$

$$Z_1 = \sigma(-0.4) = \underline{\underline{0.4013}}$$

For Z_2

$$net\ Z_2 = (-1)(-0.4) + (1)(0.4) + (1)(0.5)$$

$$net\ Z_2 = 0.4 + 0.4 + 0.5 = 1.3$$

$$Z_2 = \sigma(1.3) = \underline{\underline{0.7858}}$$

Output layer

net $y = (0.4013)(0.4) + (0.7858)(0.1) + (1)(0)$

net $y = 0.1605 + 0.0786 + 0.2$

net $y = \underline{\underline{0.4391}}$

Correct output (Before weight output)

$y = 0.6081$

3. Backward propagation

Output Error Term

$\delta y = (t-y) \cdot y(1-y)$

$\delta y = (1 - 0.6081)(0.6081)(0.3919)$

$\delta y = 0.0934$

4) Update Hidden → Output weights

$\Delta w = \alpha \delta y z$

w_{z1y}

$\Delta w = 0.25(0.0934)(0.4013)$

$= 0.00937$

$w_{z1y}^{new} = \underline{\underline{0.4094}}$

w_{z2y}

$\Delta w = 0.25(0.0934)(0.7858) = 0.01836$

$\Delta w_{z2y}^{new} = \underline{\underline{0.1184}}$

Bias to y

$\Delta b = 0.25(0.0934) = 0.02335$

$b_{y^{new}} = 0.2234$

5. Hidden layer Error Terms

$$\delta_z = z(1-z)\delta_y w$$

For Z_1

$$\delta_{z_1} = (0.4013)(0.5987)(0.0934)(0.4)$$

$$\delta_{z_1} = 0.00898$$

For Z_2

$$\delta_{z_2} = (0.7858)(0.2142)(0.0934)(0.1)$$

$$\delta_{z_2} = 0.001572$$

6. Update Input → Hidden weights

$$\Delta v = a\delta_z x$$

Updated weights for Z_1

weight	New value
V_{x1z1}	0.5978
V_{x2z1}	-0.0978
$Bias_{z_1}$	0.3023

Updated weights for Z_2

weight	New value
V_{x1z2}	-0.40039
V_{x2z2}	0.40039
$Bias_{z_2}$	0.50039

1. Network output

$y = 0.6081$

2. Updated weights

Hidden → output

- $w_{z1y} = 0.4094$
- $w_{2zy} = 0.1184$
- $b_y = 0.2234$

Input → Hidden

- $V_{x1z1} = 0.5978$
- $V_{x2z1} = -0.0978$
- $b_{z1} = 0.3023$
- $V_{x1z2} = -0.40039$
- $V_{x2z2} = 0.40039$
- $b_{z2} = 0.50039$